Advantages

1. Cheaper
2. Light weight
3. Rugged construction
4. Higher efficiency
5. Requires less maintenance

Disadvantages

1. Moderate starting torque
2. Starting torque cannot be controlled

Applications

1. Lathes
2. Drilling machines
3. Fans
4. Blowers
5. Water pumps
6. Grinders and
7. Printing machines

b) Slip Ring (or) Wound Rotor

In this type of rotor, rotor windings are similar to the stator winding. The rotor winding may be star or delta connected, distributed winding. The rotor is wound for as many poles as the member of stator poles and is always wound three phase even when the stator is wound two phase. The three phases are brought out and connected to slip rings mounted on the rotor shaft. It is possible for introduction of additional resistance in the rotor circuit during the starting period for increasing the starting torque of the motor. By varying the external resistance in the rotor circuit, the motor speed and torque can be controlled.

Advantages

1. The starting torque can be controlled by varying the rotor circuit resistance
2. The speed of the motor can also be controlled by varying the rotor circuit resistance

Disadvantages

1. Slip-ring type motor is heavier.
2. High cost
3. High rotor inertia
4. High speed limitation
5. Maintenance problems due to brushes and slip – rings

Applications

These motors are used when speed control and high starting torque is required. Mainly used in:

1. Lifts

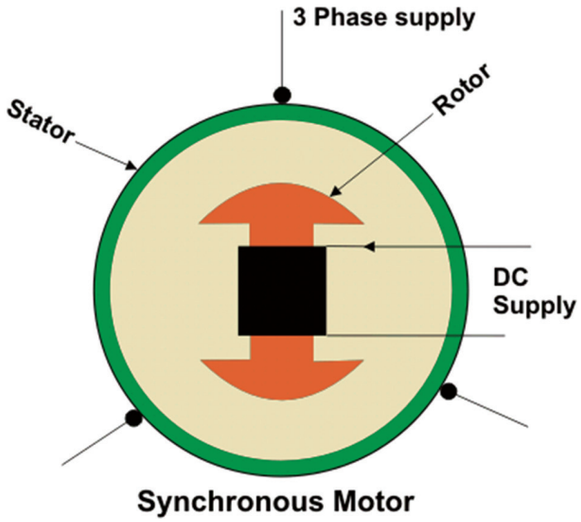


Fig 8.11 *Stator and Rotor Synchronous Motor*

2. Hoists
3. Cranes
4. Compressors

8.5.3 Slip

The difference between synchronous speed and rotor speed is called the slip speed. It is usually expressed as a percentage of synchronous speed (N_s) and represented by the symbol 'S'.

Slip Speed= $N_S - N$

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